23 April 2026

EAST WEST UNIVERSITY
Department of Computer Science and Engineering

Assignment 01, Spring 2026 Semester

Course: CSE475/ ICE478, Machine Learning, Section - 2
Instructor: Dr Raihan Ul Islam, Associate Professor, Department of CSE
Full Marks: 6

Notes:

- C2: Understand, use, and determine appropriate machine learning methods/algorithms suitable for different types of learning problems, i.e. know about their most important weaknesses and advantages.

Instructions

- This assignment is based on the STRIPS planning framework in the blocks world, as covered in Lecture 9. You are given a set of blocks placed on a table, and a robot hand that can pick up, put down, stack, and unstack one block at a time. Your task in each question is to reason about the world state, the operators that act on it, and the plans that transform an initial state into a goal state.
- Use only the four operators defined in the Operator Reference table below. Do not introduce new operators or modify the preconditions, add lists, or delete lists.
- Assume the standard blocks-world closed-world assumption and a single robot hand.
- Any action whose preconditions are not satisfied in the preceding state is invalid, regardless of the final state reached.
- Show all working. Answers without justification will receive zero even if the final plan is correct.
- Text similarity with other sources must remain below 10%. AI-generated content similarity must remain below 5%. Note: identical correct plan traces arising from the same problem will not by themselves be treated as plagiarism; the assessment will consider the working and reasoning shown.

All four blocks-world operators are reproduced below. Use exactly these definitions. Do not modify preconditions, add lists, or delete lists.

Operator	Preconditions	Effects (add / delete)
`Pickup(x)`	`clear(x), ontable(x), handempty`	`holding(x), ¬ontable(x), ¬clear(x), ¬handempty`
`Putdown(x)`	`holding(x)`	`ontable(x), clear(x), ¬holding(x), handempty`
`Stack(x,y)`	`holding(x), clear(y)`	`on(x,y), clear(x), ¬holding(x), handempty, ¬clear(y)`
`Unstack(x,y)`	`on(x,y), clear(x), handempty`	`holding(x), clear(y), ¬on(x,y), ¬clear(x), ¬handempty`

Question 2: [C2 Marks: 1+1+1 =3]

Initial state:

`{ on(b,a), ontable(a), ontable(c), ontable(d), clear(b), clear(c), clear(d), handempty }`

Goal: `{ on(a,c), on(c,d) }`

A junior student tried to solve this problem and wrote the 6-action plan shown below. The plan does not work. It has one mistake.

1. `Unstack(b, a)`
2. `Stack(b, d)`
3. `Pickup(c)`
4. `Stack(c, d)`
5. `Pickup(a)`
6. `Stack(a, c)`

Answer the following:

(a) Go through the plan one action at a time. Before each action, write down the State List at that moment. Stop when you find the action that cannot be done.

(b) Which step number is wrong? Which precondition of that action is not true at that moment? Why is that precondition not true? Explain in your own words.

(c) Imagine the wrong action is skipped and the rest of the plan is run. Will the plan reach the goal? Answer Yes or No and explain in one or two sentences why.

Question 2: [C2 Marks: 1+1+1 =3]

Initial state:

`{ ontable(a), ontable(b), ontable(c), clear(a), clear(b), clear(c), handempty }`

Goal: `{ on(a,b), on(b,c) }`

A student wrote the 8-action plan below for this problem. The plan is correct — every action can be done at its step, and at the end the goal is reached. But the plan is too long. There is a shorter way.

1. `Pickup(a)`
2. `Stack(a, c)`
3. `Unstack(a, c)`
4. `Putdown(a)`
5. `Pickup(b)`
6. `Stack(b, c)`
7. `Pickup(a)`
8. `Stack(a, b)`

(a) Show that the plan is correct. After each action, write the State List in one line.

(b) Some steps in this plan are not needed. Which step numbers can be removed without stopping the plan from reaching the goal? Explain why each of those steps is not needed.

(c) Write a shorter plan that solves the same problem. State how many actions your plan has. Explain in one or two sentences why your plan cannot be made any shorter.